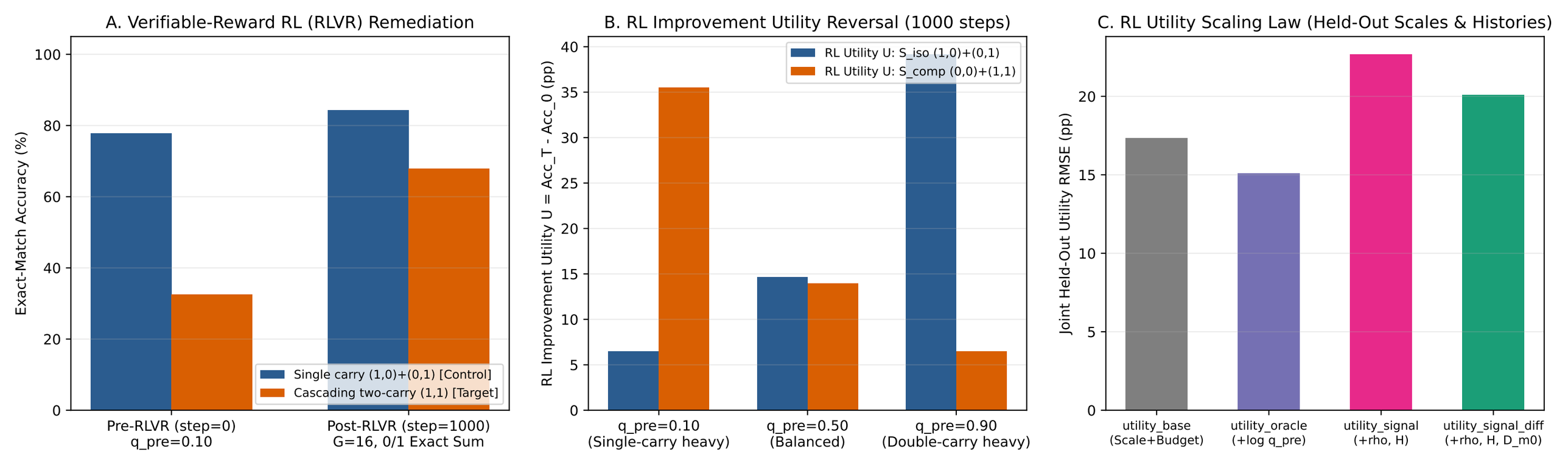
A. Verifiable-Reward RL (RLVR) Remediation
Exact-Match Accuracy (%)
0
20
40
60
80
100
Single carry (1,0)+(0,1) [Control]
Cascading two-carry (1,1) [Target]
Pre-RLVR (step=0)
q_pre=0.10
Post-RLVR (step=1000)
G=16, 0/1 Exact Sum
B. RL Improvement Utility Reversal (1000 steps)
RL Improvement Utility U = Acc_T - Acc_0 (pp)
0
5
10
15
20
25
30
35
40
RL Utility U: S_iso (1,0)+(0,1)
RL Utility U: S_comp (0,0)+(1,1)
q_pre=0.10
(Single-carry heavy)
q_pre=0.50
(Balanced)
q_pre=0.90
(Double-carry heavy)
C. RL Utility Scaling Law (Held-Out Scales & Histories)
Joint Held-Out Utility RMSE (pp)
0
5
10
15
20
utility_base
(Scale+Budget)
utility_oracle
(+log q_pre)
utility_signal
(+rho, H)
utility_signal_diff
(+rho, H, D_m0)